

Comparative Study of a Classical ML Algorithm and a Deep Learning Model

Heart Disease Risk Prediction - Logistic Regression vs. Artificial Neural Network

NCIT • Department of Computer Engineering

BE Computer Engineering, 6th Semester (BECE Day) - Course: Machine Learning - Group Case Study

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Problem Statement

- Coronary heart disease and heart attack are leading causes of death worldwide.
- Clinical testing (ECG, angiography, biomarkers) is accurate but costly and not always accessible.
- Public-health surveys (CDC BRFSS) already collect lifestyle, demographic, and self-reported health data from hundreds of thousands of respondents every year.
- Goal: use 21 self-reported indicators to flag likely heart-disease / heart-attack cases and compare a classical model against a deep learning model on the identical task.

Objectives

- Acquire and validate the BRFSS 2015 heart-disease indicators dataset for binary classification.
- Build a leakage-safe preprocessing pipeline (median imputation + standardization), shared fairly across both models.
- Implement Logistic Regression as an interpretable classical baseline.
- Implement an Artificial Neural Network (ANN) to capture non-linear feature interactions.
- Fine-tune both models and fairly compare baseline vs. tuned performance.
- Evaluate using Accuracy, Precision, Recall, F1, Specificity, ROC-AUC, and PR-AUC - not Accuracy alone.

Dataset

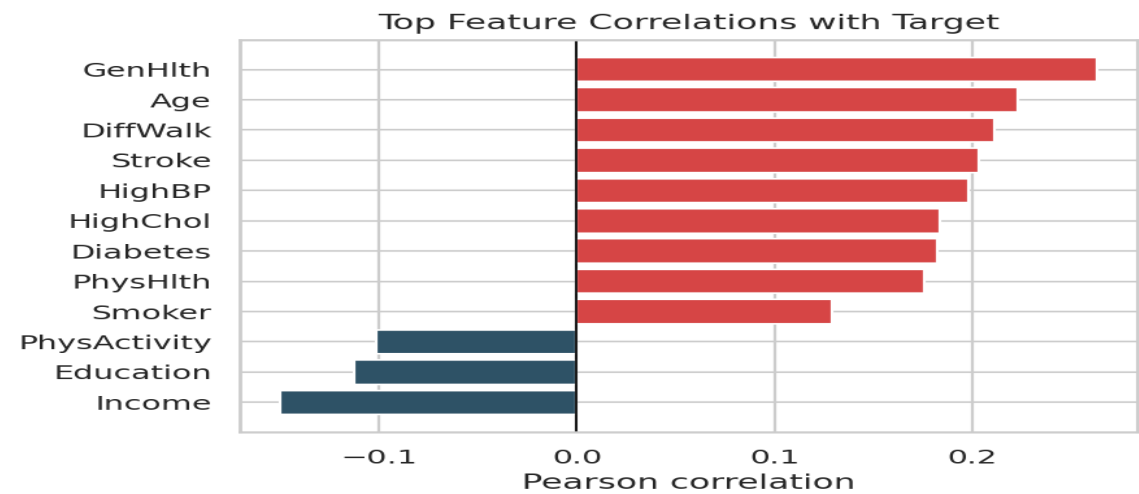
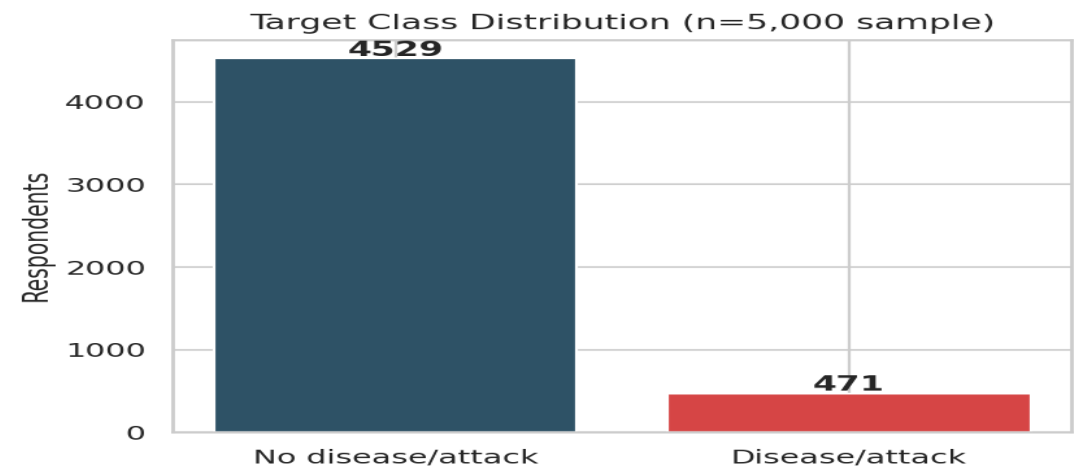
BRFSS 2015 Heart Disease Health Indicators (Kaggle / CDC)

- Full cleaned dataset: 253,680 respondents × 21 features + target (Heart Disease or Attack).
- Class balance: ~90.6% negative, ~9.4% positive - a realistic, imbalanced screening scenario.
- Working subset: stratified sample of 5,000 rows (seed = 42) for tractable Colab training.
- Split: 60% train (3,000) / 20% validation (1,000) / 20% test (1,000), all stratified.

Class	Count
No disease/attack	4,529
Disease/attack	471

Target prevalence: 9.42%

Features cover blood pressure, cholesterol, BMI, smoking, stroke, diabetes, lifestyle, healthcare access, general/mental/physical health, and demographics.



Methodology

- Schema validation & cleaning -Checks required columns, Converts values to numeric format, Removes rows without a target, Handles infinite values, Checks missing feature values.
- Stratified 5,000-row sample to stratified 60/20/20 train/validation/test split.
- Shared preprocessing: median imputation + standard scaling, fit only on training/development data.
- Train baseline Logistic Regression and baseline ANN on the training set.
- Tune Logistic Regression via 5-fold cross-validated grid search (ROC-AUC).
- Tune ANN via a structured 5-candidate search on the validation set.
- Evaluate all four models once on the same untouched test set.

Classical Model: Logistic Regression

- Models log-odds of heart disease as a linear function of standardized features to Calculates weighted sum of features and passes to sigmoid function.
- Chosen for interpretability, fast, suitable for binary classification.
- Baseline: default L2 penalty, $C = 1.0$, $\text{max_iter} = 2000$.
- Tuned via GridSearchCV (5-fold stratified CV, scoring = ROC-AUC) over:
 - Configured cross- validation : liblinear solver ,regularization penalties (L1/L2)
- Best: solver = liblinear, penalty = L1, $C = 0.1$ - CV ROC-AUC = 0.844.

Deep Learning Model: Artificial Neural Network

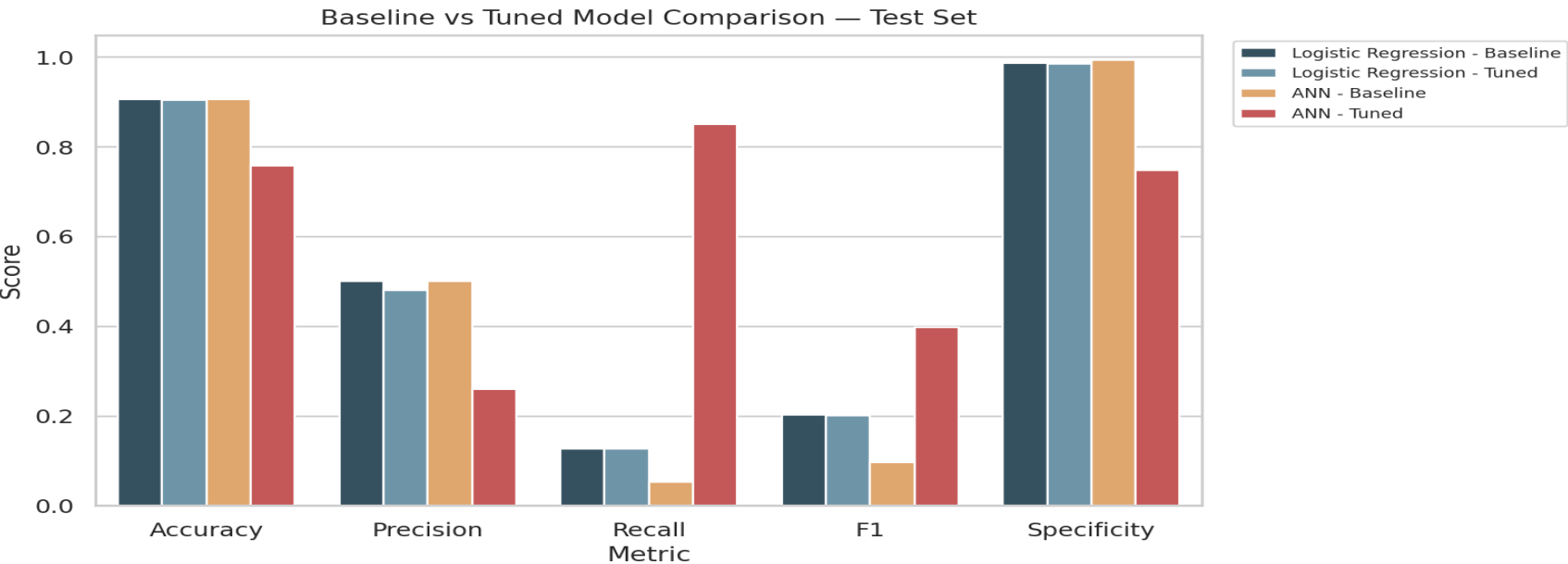
- Fully connected MLP for tabular data: Dense hidden layers (ReLU) + optional Dropout + sigmoid output.
- 21 input feature , network contains hidden layers with 128, 64 and 32 neurons
- Loss function : binary cross-entropy. Optimizer: Adam.
- Baseline: hidden layers (32, 16), no dropout, learning rate 0.001, batch size 32.
- Tuning: 5 candidates varying depth/width, dropout, learning rate, L2, batch size, class weighting - selected by validation ROC-AUC.

Configuration	Hidden layers	Val. ROC-AUC	Epoch
Small balanced	(32,16)	0.8735	12
Medium balanced	(64,32)	0.8756	4
Deep balanced (selected)	(128,64,32)	0.8839	9
Medium unweighted	(64,32)	0.8739	8
Strong regularization	(64,32,16)	0.8810	28

Results: Master Model Comparison

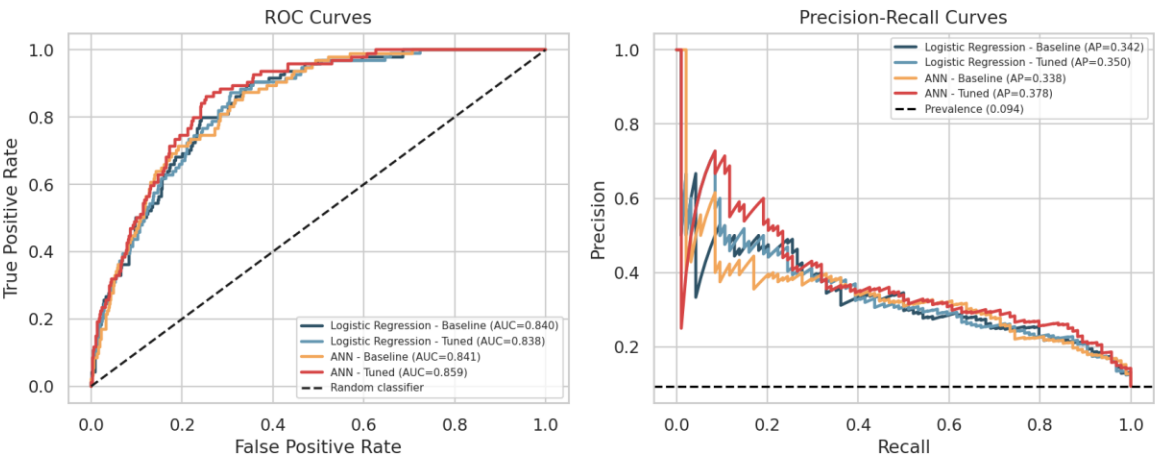
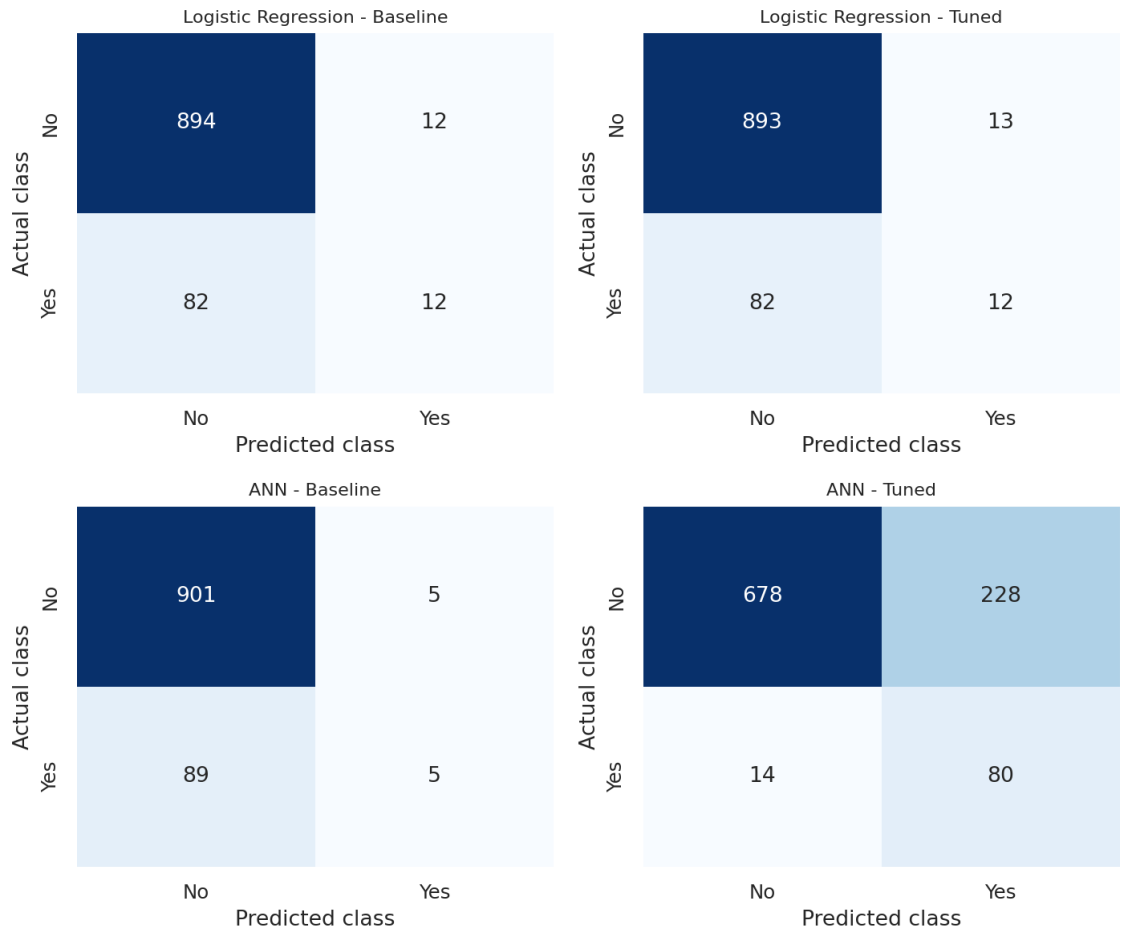
Test set, n = 1,000 (held out from all tuning)

Model	Accuracy	Precision	Recall	F1	Specificity	ROC-AUC	PR-AUC
Logistic Regression - Baseline	0.906	0.500	0.128	0.203	0.987	0.840	0.342
Logistic Regression - Tuned	0.905	0.480	0.128	0.202	0.986	0.838	0.350
ANN - Baseline	0.906	0.500	0.053	0.096	0.994	0.841	0.338
ANN - Tuned	0.758	0.260	0.851	0.398	0.748	0.859	0.378



Confusion Matrices, ROC & PR Curves

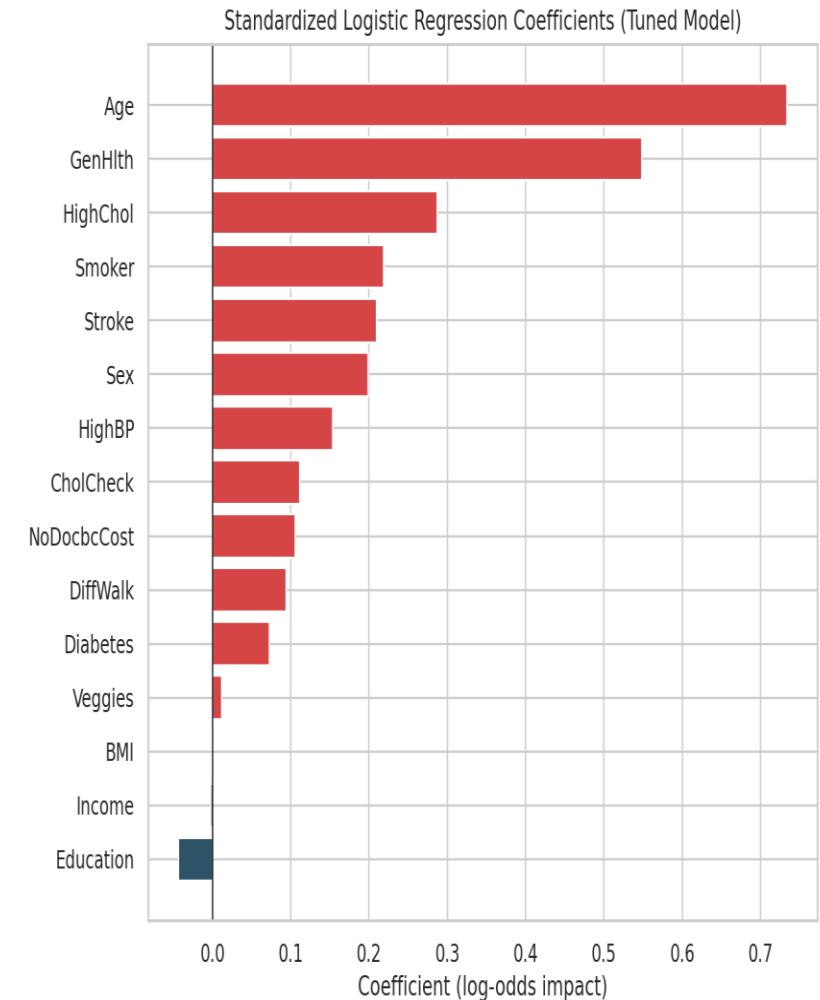
Test-Set Confusion Matrices (n=1,000)



Tuned ANN: best ROC-AUC (0.859) and best PR-AUC (0.378) - finds far more true positives (80 vs. 12) at the cost of more false alarms (228 vs. 12–13).

Key Findings & Discussion

- Both baselines had ~90.6% Accuracy but very low Recall (5–13%) - they largely ignored the rare positive class.
- Tuned Logistic Regression (L1, C=0.1) auto-selected features, zeroing 6 of 21 coefficients, but Recall stayed low (best CV config did not use class weighting).
- Class-weighting the ANN was the turning point: Recall jumped from 5.3% to 85.1%, and ROC-AUC rose to 0.859 - the best of all four models.
- Trade-off: the tuned ANN finds many more true heart-disease cases, but at much lower precision/specificity - a classic imbalanced-screening trade-off.
- Strongest linear risk predictors: Age, General Health, High Cholesterol, Smoking, Stroke history, and Sex.



Optimization Options & Conclusion

- Optimize the decision threshold (currently 0.50) directly on the validation set instead of a fixed cut-off.
- Apply class-weighting/resampling (SMOTE - Synthetic Minority Over Sampling Technique) to Logistic Regression too, not only the ANN.
- Use automated search (KerasTuner / Optuna) and add ensemble baselines (Random Forest, XGBoost).
- Train on the full 253,680-row dataset to give the ANN far more positive examples to learn from.
- Conclusion: model choice depends on the goal - tuned ANN maximizes detection (Recall/ROC-AUC); tuned Logistic Regression stays the transparent, conservative choice.

GitHub Repo: https://github.com/adityashah01/HeartDisease_Prediction



Thank You

-Any Queries?